

Heaven's Light is Our Guide



Rajshahi University of Engineering and Technology

Department of Electrical and Computer Engineering

Report 02

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Submitted by
Redwanul huq Tamim
2410023

Submitted to
Mst. Mazeda Noor Tasnim
Lecturer, ECE, RUET

Problem 1: Boolean Simplification

Problem Definition

The purpose of this problem is to simplify the Boolean expression:

$$F(A, B, C) = A'BC + AB'C + ABC' + ABC$$

using Boolean algebra and a Karnaugh map (K-map). After finding the simplified expression, both the original and simplified circuits will be built in Logisim and simulated to verify that they produce the same output.

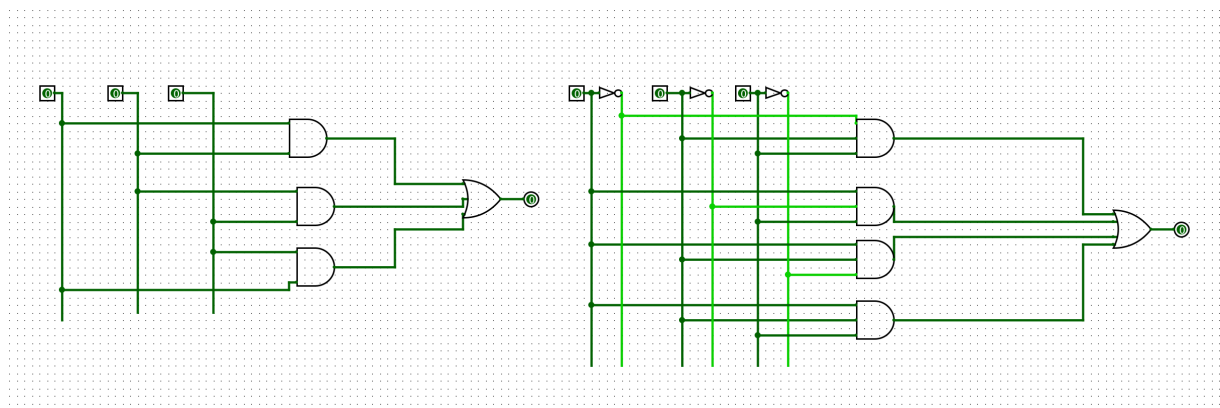
Objective

The main objective is to reduce the given three-variable sum-of-minterms Boolean expression into its simplest Sum-of-Products (SOP) form. The simplification will first be done using Boolean algebra and then verified using a Karnaugh map. Finally, both the original and simplified circuits will be designed in Logisim and tested to confirm that they work identically.

Algebraic Simplification

$$\begin{aligned} F(A, B, C) &= A'BC + AB'C + ABC' + ABC \\ &= A'BC + ABC + AB'C + ABC + ABC' + ABC \\ &= BC(A' + A) + AC(B' + B) + AB(C' + C) \\ &= BC + AC + AB \\ &= AB + AC + BC \end{aligned}$$

Circuit Diagram:



Truth Table:

A	B	C	$F = AB + AC + BC$
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

Karnaugh Map

A	BC			
	00	01	11	10
0	0	0	1	0
1	0	1	1	1

Table 2: K-map for $F(A, B, C) = A'BC + AB'C + ABC' + ABC$

K-map Simplification

Three pairs of adjacent 1-cells can be formed in the Karnaugh map. The pair at $BC = 11$ across both rows gives the term BC . The pair where $A = 1$ and $C = 1$, covering the columns $BC = 01$ and 11 , gives AC . Similarly, the pair where $A = 1$ and $B = 1$, covering $BC = 11$ and 10 , gives AB .

Therefore, the simplified expression is:

$$F = AB + AC + BC$$

This result is the same as the expression obtained using Boolean algebra.

Discussion

The original circuit directly implements the given expression. It requires four 3-input AND gates, one for each product term, along with inverters for generating A' , B' , and C' where necessary. The outputs of the AND gates are then connected to a 3-input OR gate.

The simplified circuit is much simpler. It uses only three 2-input AND gates to generate AB , AC , and BC , followed by an OR gate. No inverters are needed.

Both circuits were tested in Logisim using the same input combinations. The outputs of the two circuits were identical for all possible inputs. This confirms that the Boolean algebra and K-map simplifications are correct. The simplified circuit is also more efficient because it uses fewer gates, fewer gate inputs, and does not require any inverters.

Conclusion

The given Boolean expression $F = A'BC + AB'C + ABC' + ABC$ was successfully simplified to $F = AB + AC + BC$ using Boolean algebra and a Karnaugh map. The original and simplified circuits were also tested in Logisim, and both produced the same output for every input combination. Therefore, the simplification was successfully verified.

Problem 2: Absorption Law

Problem Definition

Simplify the Boolean expression:

$$F(A, B, C) = A(A + B)(A + B + C)$$

using Boolean algebra and a Karnaugh map. The original and simplified circuits will then be built and tested in Logisim to verify that they are equivalent.

Objective

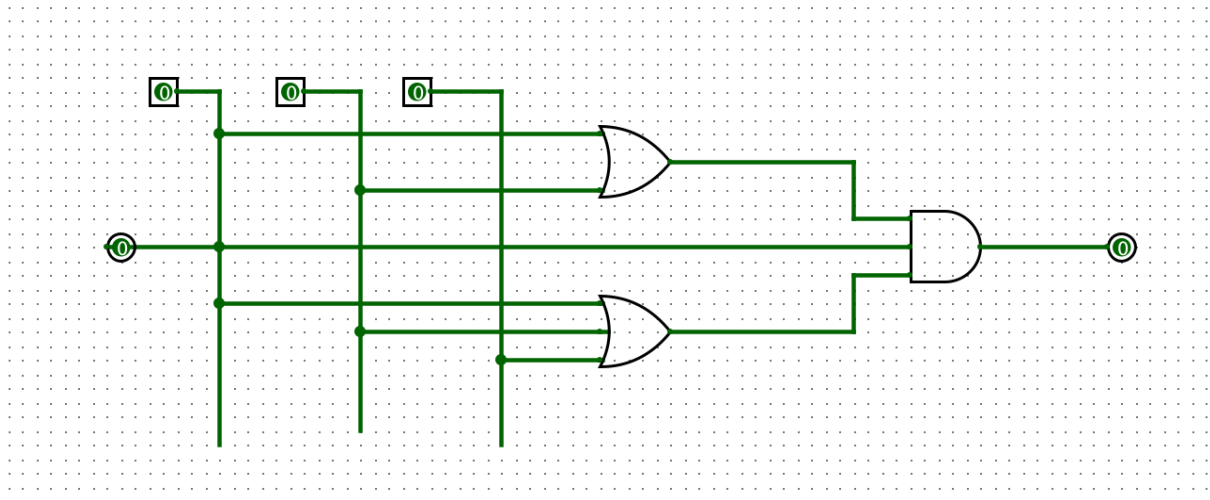
The objective is to simplify the given three-variable product-of-sums expression to its simplest form using Boolean algebra and a Karnaugh map. The simplified result will then be verified by comparing the original and simplified circuits through Logisim simulation.

Algebraic Simplification

$$\begin{aligned} F(A, B, C) &= A(A + B)(A + B + C) \\ &= (A + AB)(A + B + C) \\ &= A(A + B + C) + AB(A + B + C) \\ &= A + AB + AC + AB + AB + ABC \\ &= A + AB + AC + ABC \end{aligned}$$

$$= A(1 + B + C + BC)$$

$$= A$$



Truth Table:

A	B	C	F = A
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

Karnaugh Map

	<i>BC</i>			
<i>A</i>	00	01	11	10
0	0	0	0	0
1	1	1	1	1

Table 4: K-map for $F(A, B, C) = A(A + B)(A + B + C)$

All four cells in the $A = 1$ row are 1, while the entire $A = 0$ row is 0. These four adjacent cells form a single group spanning the whole row, eliminating both B and C and leaving only $F = A$, exactly matching the algebraic result.

Discussion

The original circuit builds $A + B$ using one OR gate, extends it to $A + B + C$ with a second OR gate, and ANDs both intermediate results with A itself. Since A is common to every term, all higher-order terms are absorbed and the entire network collapses to the single literal A , so the

simplified circuit is nothing more than a direct wire from input A to the output, with no logic gates involved. This is an example of the absorption law $A + AB = A$ applied twice in succession. Testing both circuits in Logisim across all combinations of A, B, and C showed that the output always equals A regardless of B and C, confirming the simplification.

Conclusion

The expression $F = A(A + B)(A + B + C)$ was shown to reduce to $F = A$ through repeated application of the absorption law, a result confirmed by both the Karnaugh map and Logisim simulation, which demonstrated that B and C have no effect whatsoever on the output.

Problem 3: De Morgan's Theorem

Problem Definition

Simplify the Boolean expression $F(A, B, C) = (A + (BC)')(AB + ABC)$ using Boolean algebra and the Karnaugh map, and verify both the original and simplified circuits through logic simulation.

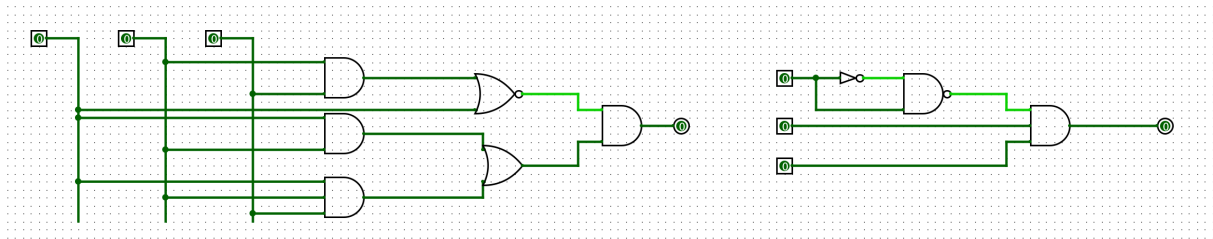
Objective

To reduce a mixed AND-OR-complement expression using De Morgan's theorem and Boolean algebra, to confirm the result with a Karnaugh map, and to verify in Logisim that the simplified circuit behaves as a constant.

Algebraic Simplification

$$\begin{aligned} F(A, B, C) &= (A + (BC)')(AB + ABC) \\ &= A' \cdot BC \cdot (AB + ABC) && \text{(De Morgan's theorem)} \\ &= A'BC \cdot AB(1 + C) \\ &= A'BC \cdot AB \\ &= (AA')BCB \\ &= 0 \end{aligned}$$

Circuit Diagram:



Truth Table:

A	B	C	$F = 0$
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	0

Karnaugh Map

	BC			
A	00	01	11	10
0	0	0	0	0
1	0	0	0	0

Table 6: K-map for $F(A, B, C) = (A + (BC)')'(AB + ABC)$

Every cell of the map contains a 0, so there are no 1-cells available for grouping. A Karnaugh map with no populated cells represents the constant function $F = 0$, confirming the algebraic result.

Discussion

Applying De Morgan's theorem to $(A + (BC)')'$ converts it into $A' \cdot BC$, after which the second factor $AB + ABC$ simplifies to AB by absorption ($AB + ABC = AB(1 + C) = AB$). Multiplying $A'BC$ by AB introduces the term $A \cdot A'$, which is always 0, forcing the entire expression to 0 regardless of the values of A , B , and C . This was confirmed in Logisim by cycling through all

eight input combinations on the original circuit and observing that the output never leaves logic 0, exactly as predicted by the all-zero Karnaugh map.

Conclusion

The expression $F = (A+(BC)')'(AB+ABC)$ was shown to be identically zero for every input combination. This was verified algebraically, confirmed by an empty Karnaugh map, and validated through Logisim simulation, which produced a constant logic-0 output regardless of A, B, and C.

Problem 4: Two-Variable Reduction

Problem Definition

Simplify the Boolean expression $F(A, B) = (B'(A + B) + (A + B)(A + B'))B'$ using Boolean algebra and the Karnaugh map, and verify both the original and simplified circuits through logic simulation.

Objective

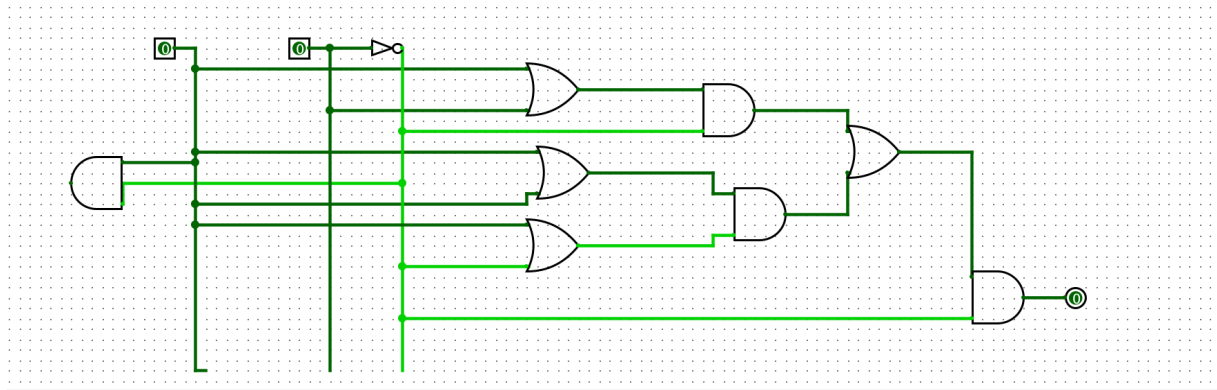
To reduce a two-variable expression built from nested OR and AND terms to its minimal form, to confirm this reduction with a Karnaugh map, and to verify the equivalence of the original and simplified circuits in Logisim.

Algebraic Simplification

$$\begin{aligned} F(A, B) &= (B'(A + B) + (A + B)(A + B'))B' \\ &= (AB' + BB' + AA + AB' + AB' + BB')B' \\ &= (A + AB' + AB' + AB')B' \\ &= A(1 + B' + B' + B')B' \\ &= AB' \end{aligned}$$

.

Circuit Diagram:



Truth Table:

A	B	$F = AB'$
0	0	0
0	1	0
1	0	1
1	1	0

Karnaugh Map

A	B	
	0	1
0	0	0
1	1	0

Table 8: K-map for $F(A, B) = (B'(A + B) + (A + B)(A + B'))B'$

Only a single cell, at $A = 1, B = 0$, is populated. A lone 1-cell cannot be combined with any neighboring cell, so it is read directly as the product of its coordinate literals, giving $F = AB'$, which matches the algebraic result.

Discussion

When the original expression is expanded, several terms are produced, including AB' , A , BB' , and AA . Since $BB' = 0$ and $AA = A$, the expression becomes much simpler. After simplifying and combining the remaining terms, the final result is AB' .

Therefore, the original circuit, which uses multiple AND, OR, and NOT gates, is functionally equivalent to a simple 2-input AND gate with B inverted. Both circuits were tested in Logisim

for all four possible input combinations of A and B. In every case, the outputs were the same. The output becomes 1 only when $A = 1$ and $B = 0$, which matches the simplified expression $F = AB'$.

Conclusion

The Boolean expression $F = (B'(A + B) + (A + B)(A + B'))B'$ was successfully simplified to $F = AB'$ using Boolean algebra and verified using a Karnaugh map. The original and simplified circuits were also tested in Logisim for all possible input combinations. Both circuits produced identical outputs, confirming that the simplification is correct.

References

1. M. M. Mano, Digital Design, 5th ed., Upper Saddle River, N.J.: Pearson, 2013.
2. R. J. Tocci, N. S. Widmer, and G. L. Moss, Digital Systems: Principles and Applications, 11th ed., Upper Saddle River, N.J.: Pearson, 2011.